Bayesian Optimisation of a Bio-Based Polyester Resin for Cured Thermoset Strength

7th July 2025

Thomas Højlund-Dodd, James Elliott

Abstract

This paper will focus on particular group of thermosets, namely, polyester resins. These liquid resins generally cured to form the matrix that surrounds reinforcing materials within the structure of composite materials such as glass-reinforced plastic. This paper initially elucidates the method by which an entirely bio-based polyester resin was manufactured, from simple organic acids and alcohols. Subsequently, a description of the method and associated results of Bayesion optimisation routines designed to improve the strengths exhibited by these materials will be brought to bear. The most salient findings include the mere possibility that these prepolymers can be manufactured simply and cheaply. Additionally, it is worth noting their strength being comparable after optimisation to their contemporary incumbents; the styrene-based unsaturated polyester resins.

1 Introduction

Plastics encompass a plethora of polymeric materials which provide a wide-range of desirable material properties; and given this, have seen use in a variety of applications, spanning both commodity and engineering settings [5].

Since the 1950s, the annual mass of plastics manufactured has grown from 2 Mt yr^{-1} [4] to around 460 Mt yr^{-1} in 2019 [2]; naturally, plastic waste tracked this rise.

In 2019, 353 Mt of plastic waste was generated globally, with 49% landfilled, 22% mismanaged, 29% incinerated, and 9% recycled [2].

Mismanaged plastics may degrade and pollute watercourses with microplastics [7], incinerated plastics provide a means by which fossil carbon may reach the atmosphere [3], and sluggish development of recycling globally makes this the least likely fate of waste plastic [2]; these are a subset of the many challenges presented by use of this family of materials.

Since around 85% of virgin plastics manufactured today can be classified as thermoplastic [10], a dominant proportion of literature is focussed upon tackling the various associated challenges;

This paper will focus upon the lesser-known grouping of thermosets given the unique challenges and opportunities associated with these materials.

Polyester resins are used for a number of reasons, including ease of handling in liquid form, rapid curing, and excellent dimensional stability [8].

Generally thermosets are either landfilled [1] or incinerated [9].

Many thermosets exist including (and not limited to) phenolic resins, unsaturated polyester resins, epoxy resins, and polyurethanes [6].

2 Method

3 Results

4 Conclusion

5 Bibliography

[1] Huaguo Chen et al. “Recycling thermoset plastic waste for manufactur-

ing green cement mortar". In: *Cement and Concrete Composites* 137.104922 (2023).

[2] OWID (Our World in Data). *Plastic Pollution*. Ed. by Hannah Ritchie et al. Retrieved July 12th, 2025. 2025. URL: `https://ourworldindata.org/plastic-pollution`.

[3] Orla Eriksson et al. "Plastic waste as a fuel - CO2-neutral or not?" In: *Energy and Environmental Science* 2 (2009), pp. 907–914.

[4] Roland Geyer et al. "Production, use, and fate of all plastics ever made". In: *Science Advances* 3.e1700782 (2017).

[5] Marianne Gilbert. "Brydson's Plastics Materials". In: ed. by Marianne Gilbert. Eighth. 2017. Chap. 1.

[6] Sidney Goodman. *Handbook of Thermoset Plastics*. Ed. by Sidney Goodman. First. Noyes Publications, 1986.

[7] Hiroshan Hettiarachchi et al. "Microplastic Pollution Prevention- The Need for Robust Policy Interventions to Close the Loopholes in Current Waste Management Practices". In: *International Journal of Environmental Research and Public Health* 20.14 (2023).

[8] John Lawrence. *Polyester Resins*. Ed. by Herbert Simonds. First. Reinhold Publishing Corporation, 1960.

[9] Eva Topham. "Recycling offshore wind farms at decommissioning stage". In: *Energy Policy* 129 (2019), pp. 698–709.

[10] Chunyan Wang et al. "Critical review of global plastics stock and flow data". In: *Journal of Industrial Ecology* 25.5 (2021).